

 L15_L17.pdf

Example Lecture Scribe (structure only):

Lecture 0: Introduction to Logic

1. Definitions

Proposition: A declarative statement that is either true or false, but not both.

Notation: Denoted by lowercase letters p, q, r .

Compound Proposition: A proposition formed from existing propositions using logical operators.

2. Logical Operators

Negation ($\neg p$): The statement "it is not the case that p ".

Truth Table: If p is True, $\neg p$ is False.

3. Theorems and Proofs

Theorem 1 (De Morgan's Law): $\neg(p \wedge q) \equiv \neg p \vee \neg q$

Proof:

Construct a truth table for both sides.

Observe that for all truth value assignments of p and q , the column for $\neg(p \wedge q)$ is identical to $\neg p \vee \neg q$.

Thus, the statements are logically equivalent.

4. Worked Example

Problem: Show that $p \vee \neg p$ is a tautology.

Solution:

Step 1: Assign logical values.

Step 2: If $p=T$, then $\neg p=F$, so $T \vee F = T$.

Step 3: If $p=F$, then $\neg p=T$, so $F \vee T = T$.

Conclusion: In all cases, the result is True.

Instructions:

Using the same structure as the example above, generate a lecture scribe for the provided context (lecture slides/textbook).

Requirements:

Purpose: The scribe must serve as exam reference material.

Content: Include definitions, notation, assumptions, theorem statements, proofs, derivations, and worked examples exactly as presented in the lecture materials.

Faithfulness: Do not introduce new examples, creative explanations, or intuition not discussed in the context. Everything must be traceable to the provided files.

Format: Use clear Markdown headers, bullet points for definitions, and for all mathematical notation.

dont give the latex code just the notes



Here is the lecture scribe based exactly on the provided materials.

Lecture Scribe: Joint Random Variables + Joint Distributions + Examples

[cite_start]**Course:** CSE400 - Fundamentals of Probability in Computing [cite: 1]

[cite_start]**Instructor:** Dhaval Patel, PhD [cite: 4] [cite_start]**Date:** March 10, 2026 [cite: 7]

1. Introduction to Joint Random Variables

- [cite_start]**Motivation:** In science and real life, we are often interested in two or more random variables at the same time[cite: 30, 43].
- [cite_start]**Purpose:** The random variables have a joint distribution that allows us to compute probabilities of events involving both variables and understand the relationship between them[cite: 82].
- [cite_start]**Dependence:** This is simplest when the variables are independent[cite: 83].
[cite_start]When they are not, covariance and correlation are used as measures of the nature of the dependence between them[cite: 84].
- **Examples:**
 - [cite_start]Height and weight of giraffes[cite: 32].
 - [cite_start]Frequency of exercise and rate of heart disease[cite: 34].

- [cite_start]Smart Transportation: X = Vehicle Speed and Y = Traffic Density[cite: 46, 47].
- [cite_start]Wireless Network Performance: X = Signal-to-Noise Ratio (SNR) and Y = Data Throughput[cite: 51]. [cite_start]Throughput depends on SNR, making X and Y correlated random variables[cite: 52, 56].
- [cite_start]Weather Prediction: X = Rainfall and Y = Temperature[cite: 68].

2. Discrete Case: Joint Probability Mass Function (PMF)

- [cite_start]**Setup:** Suppose X and Y are discrete random variables where $X \in \{x_1, x_2, \dots, x_n\}$ and $Y \in \{y_1, y_2, \dots, y_m\}$ [cite: 103, 104].
- [cite_start]**Product Set:** The ordered pair (X, Y) takes values in the product set: $\{(x_1, y_1), (x_1, y_2), \dots, (x_n, y_m)\}$ [cite: 105, 106].
- [cite_start]**Definition (Joint PMF):** $p(x_i, y_j) = P(X = x_i, Y = y_j)$ [cite: 107, 108].
[cite_start]This gives the probability of the joint outcome occurring simultaneously[cite: 109].
- **Properties:**
 - [cite_start] $0 \leq p(x_i, y_j) \leq 1$ [cite: 144].
 - [cite_start] $\sum_{i=1}^n \sum_{j=1}^m p(x_i, y_j) = 1$ [cite: 145].
- [cite_start]**Example (Rolling Two Dice):** * **Experiment:** Roll two fair dice[cite: 175]. [cite_start]Let X = value on the first die, and T = total (sum) of both dice[cite: 176, 177, 178].
 - [cite_start]Since each outcome (i, j) has probability $\frac{1}{36}$, we can construct the joint probability table[cite: 188, 189, 190].
 - [cite_start]**Observations:** Entries in the table are either 0 or $\frac{1}{36}$, and X and T are not independent[cite: 203, 204].

3. Continuous Case: Joint Probability Density Function (PDF)

- [cite_start]**Analogy:** The continuous case is analogous to the discrete case, where discrete values become continuous intervals, Joint PMF becomes Joint PDF, and sums become double integrals[cite: 220, 221, 222].
- [cite_start]**Setup:** If $X \in [a, b]$ and $Y \in [c, d]$, then the pair (X, Y) takes values in the product region: $[a, b] \times [c, d]$ [cite: 233].
- [cite_start]**Definition (Joint PDF):** A function $f(x, y)$ is called the joint probability density function of X and Y if $P((X, Y) \text{ lies in a small rectangle of area } dx \, dy) \approx f(x, y) dx \, dy$ [cite: 234, 235, 236].
- **Key Properties:**
 - [cite_start] $f(x, y) \geq 0$ [cite: 283].

- [cite_start] $\int_c^d \int_a^b f(x, y) dx dy = 1$ [cite: 284].
- [cite_start]The joint PDF $f(x, y)$ may take values greater than 1. It is a probability density, not a probability[cite: 285].

4. Worked Example: Continuous Joint PDF

- [cite_start]**Problem:** Given $f(x, y) = 4xy$, where $0 \leq x \leq 1$ and $0 \leq y \leq 1$ [cite: 291, 292].
[cite_start]Jointly, (X, Y) takes values in the unit square $[0, 1]^2$ [cite: 298].
- [cite_start]**Task 1: Show that $f(x, y)$ is a valid joint PDF.**[cite: 294].
 - [cite_start]**Step 1:** Check Validity (Normalization)[cite: 326].
 - [cite_start] $\int_0^1 \int_0^1 4xy dx dy = \int_0^1 (\int_0^1 4xy dx) dy = \int_0^1 2y dy = [y^2]_0^1 = 1$ [cite: 334].
 - [cite_start]**Conclusion:** Hence $f(x, y) \geq 0$ and total probability is 1. So it is a valid joint PDF[cite: 335].
- [cite_start]**Task 2: Compute $P(A)$ for the event $A = \{X < 0.5, Y > 0.5\}$.**[cite: 296, 297].
 - [cite_start]**Step 2:** Compute $P(A)$ [cite: 344].
 - [cite_start] $P(A) = \int_0^{0.5} \int_{0.5}^1 4xy dy dx = \int_0^{0.5} [2xy^2]_{0.5}^1 dx = \int_0^{0.5} 2x(1 - 0.25) dx = \int_0^{0.5} \frac{3}{2}x dx = [\frac{3}{16}]$ [cite: 345].

5. Joint Cumulative Distribution Function (Joint CDF)

- [cite_start]**Notation:** We use the notation $\{X \leq x \text{ and } Y \leq y\}$ to mean the event $X \leq x, Y \leq y$ [cite: 368, 370, 371].
- [cite_start]**Definition:** The joint cumulative distribution function is defined as $F(x, y) = P(X \leq x, Y \leq y)$ [cite: 372, 373].
- [cite_start]**Continuous Case:** * If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$, the joint CDF is obtained by integrating the density[cite: 390].
 - [cite_start] $F(x, y) = \int_c^y \int_a^x f(u, v) du dv$ [cite: 391].
 - [cite_start]**Recovering the Joint PDF:** Since there are two variables, we use partial derivatives: $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$ [cite: 400, 401].
- **Discrete Case:**
 - [cite_start]If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, the joint CDF is obtained by summing probabilities[cite: 412, 413].
 - [cite_start] $F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$ [cite: 428].
 - [cite_start]**Interpretation:** Add all probabilities in the table for rows with $x_i \leq x$ and columns with $y_j \leq y$ [cite: 429, 430, 431, 432].

- **Properties of the Joint CDF:** The joint CDF $F(x, y)$ of X and Y must satisfy several properties[cite: 443]:
 - $F(x, y)$ is non-decreasing. [cite_start]If x or y increase, then $F(x, y)$ must stay constant or increase[cite: 445].
 - $F(x, y) = 0$ at the lower-left of the joint range. [cite_start]If the lower-left corner is $(-\infty, -\infty)$, then $\lim_{(x,y) \rightarrow (-\infty, -\infty)} F(x, y) = 0$ [cite: 454, 455].
 - $F(x, y) = 1$ at the upper-right of the joint range. [cite_start]If the upper-right corner is (∞, ∞) , then $\lim_{(x,y) \rightarrow (\infty, \infty)} F(x, y) = 1$ [cite: 466, 467].
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Would you like me to extract the data from the "SNR vs Throughput" table into a proper Markdown table so you can review it alongside these notes?